

Uber Travel Behavior Analysis

Executive Portfolio Report

Executive Summary

This project analyzes 653 cleaned Uber trips from 2016 to uncover travel behavior patterns, trip purposes, temporal demand trends, and location-based mobility insights. The analysis shows that Uber usage was overwhelmingly business-oriented, with demand concentrated around specific hours, weekdays, and trip purposes.

Business Question

How can Uber trip data be used to understand customer travel behavior and identify operational opportunities?

Dataset Overview

Original dataset: 1,156 trips. After removing records with missing values, 653 trips remained for analysis. Features include trip dates, category, locations, mileage, and trip purpose.

Methodology

Data cleaning, datetime conversion, feature engineering (hour, day, weekday, month), exploratory data analysis, and visualization-based insight generation.

Key Finding 1: Business Dominates Usage

The overwhelming majority of trips were categorized as business travel, indicating strong professional and work-related usage patterns.

Key Finding 2: Peak Travel Periods

Trip activity was concentrated within specific hours of the day, revealing predictable demand windows that can support driver allocation and demand forecasting.

Key Finding 3: Purpose-Driven Mobility

Meetings, customer visits, meals/entertainment, and errands emerged as major reasons for travel, showing that rides were often linked to business objectives.

Key Finding 4: Weekly and Monthly Trends

Trip volumes varied across weekdays and months, suggesting recurring business cycles rather than random travel behavior.

Key Finding 5: Mobility Hubs

A small number of origin locations generated a disproportionately large share of trips, indicating concentrated demand centers.

Business Recommendations

Develop stronger corporate travel programs, optimize driver deployment around peak demand periods, improve location-based resource allocation, and support demand forecasting initiatives.

Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook.